IE 7615 Discriminative Deep Learning Project

Final Project Report: Single Object Identification and Multi Object Detection

Group 2

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1. Project Overview

Every student in the class photographed one unique object under different lighting, angles and backgrounds. This gives 73 Object IDs (OBJ001 to OBJ073). The goal of the project is a system that can recognise these objects. The work was done in three milestones, and each milestone builds on the one before it.

In Milestone 1 we classified a photograph that shows one object. In Milestone 2 we moved to multi object detection, where the model must return the Object ID and the bounding box of every object in one image. In Milestone 3 we trained the final detector on more data, evaluated it once on the held out test set, analysed its errors and prepared a live demo. Table 1 summarises the three milestones.

Milestone	Task	Final model	Main test result
1	Identify the single object in a photograph (classification)	EfficientNetB0 with two stage fine tuning	Accuracy 97.4%, macro F1 0.974
2	Identify and locate every object in a multi object image (detection)	YOLOv8s on grid images (800 training grids, 50 epochs)	mAP@0.50 0.909, mAP@0.50:0.95 0.850
3	Final detector, final evaluation, error analysis and live demo	YOLOv8s on grid images (1,200 training grids, 60 epochs)	mAP@0.50 0.912, mAP@0.50:0.95 0.856, precision 0.919, recall 0.883

Table 1. Summary of the three milestones. The final model is highlighted.

Our group's objects are OBJ036 (Pencil Case) and OBJ038 (Perfume Bottle). OBJ036 is followed through every stage of the report.

2. Dataset and Data Preparation

2.1 Source photographs

Each object has its own folder, and the folder name is the class label. The shared collection contains 6,964 object photographs, with 90 to 106 photographs per Object ID (mean 95.4). The classes are close to balanced, so no class weighting was needed. The 379 background only photographs (files marked _bg_) contain no object and were not used in any milestone. All photographs were resized to 224 × 224 pixels.

2.2 Split used in Milestone 1

For Milestone 1 the photographs were split 70 / 15 / 15 into training, validation and test sets, stratified by Object ID with a fixed seed (42). Every Object ID appears in all three sets in the same proportions, and every model was trained and evaluated on the same photographs. The test set was used only for final reporting.

2.3 Cleaning added for Milestones 2 and 3

Before building multi object images, the photographs were cleaned in code. OBJ054_017 was excluded because it is not a photo of the object. Exact duplicates were found by comparing the decoded pixels, and 16 extra copies were removed (in OBJ011, OBJ022, OBJ023, OBJ028, OBJ029, OBJ062 and OBJ066),

keeping one copy of each. Every photograph was rotated according to its EXIF orientation and resized to 224 × 224, which fixed the two photographs with a different size (OBJ003_072 at 224 × 192 and OBJ023_001 at 225 × 224). This leaves 6,947 unique photographs.

The unique photographs were split again 70 / 15 / 15, stratified by Object ID with seed 42. The split was made before any grid image was generated, and a check in the notebook confirms that no photograph, or a copy of it, appears in more than one split. Table 2 compares the two splits.

Split	Milestone 1 photographs	Milestones 2 and 3 unique photographs	Share
Train	4,874	4,862	70%
Validation	1,045	1,042	15%
Test	1,045	1,043	15%
Total	6,964	6,947	100%

Table 2. Photographs per split in each milestone.

3. Milestone 1: Single Object Classification

3.1 Models tested

We compared one CNN trained from scratch with four CNNs pretrained on ImageNet (Table 3). All models were built in TensorFlow 2.20 and Keras 3 and trained on a Google Colab T4 GPU.

Model	Type	Params (M)	Classification head	Layers unfrozen in Stage 2
Custom CNN	From scratch	1.61	GAP, Dropout 0.4, Dense	All layers trained
VGG16	Transfer learning	14.75	GAP, Dropout 0.3, Dense	block5
ResNet50	Transfer learning	23.74	GAP, Dropout 0.3, Dense	conv5 stage
MobileNetV2	Transfer learning	2.35	GAP, Dropout 0.3, Dense	From block_13
EfficientNetB0	Transfer learning	4.14	GAP, Dropout 0.3, Dense	From block6a

Table 3. Models compared in Milestone 1. GAP = global average pooling; Dense = 73 way softmax output.

3.2 Training

Custom CNN. Five convolution blocks (3 × 3 convolution, batch normalization, ReLU and max pooling) with 32, 64, 128, 256 and 512 filters, followed by global average pooling, dropout (0.4) and a softmax layer. It was trained end to end for up to 60 epochs with Adam (learning rate 0.001), early stopping on validation accuracy (patience 10), and a schedule that halves the learning rate when validation loss stops improving for 4 epochs.

Transfer learning models. Each pretrained backbone used its original ImageNet preprocessing and was trained in two stages. In Stage 1 the backbone was frozen and only the new head was trained (Adam, learning rate 0.001, up to 15 epochs). In Stage 2 the top block(s) of the backbone were unfrozen and trained together with the head at a learning rate of 0.0001 for up to 15 more epochs. Batch

normalization layers stayed frozen, which keeps fine tuning stable on a small dataset. Both stages used early stopping on validation accuracy (patience 4).

Training images were augmented on the fly with random horizontal flips, rotation (±8%), zoom (±15%), translation (±10%), brightness (±20%) and contrast (±20%). All models used batch size 32 and sparse categorical cross entropy loss. For every model, the checkpoint with the highest validation accuracy was kept for testing.

3.3 Results

Table 4 summarises the results. Models are ranked by validation accuracy, which was the selection criterion. Test accuracy, top 5 accuracy and macro F1 are reported on the held out test set.

Model	Val acc	Test acc	Top 5	Macro F1	Params (M)	Size (MB)	Epochs	Train (min)
EfficientNetB0	98.2%	97.4%	99.7%	0.974	4.14	16.6	30	9.9
ResNet50	97.5%	97.0%	99.6%	0.970	23.74	94.9	28	16.5
MobileNetV2	96.2%	96.2%	99.3%	0.962	2.35	9.4	29	7.1
VGG16	95.5%	95.0%	98.7%	0.950	14.75	59.0	23	13.8
Custom CNN	77.0%	78.5%	92.6%	0.785	1.61	6.4	55	23.2

Table 4. Milestone 1 results (test set: 1,045 photographs). The selected model is highlighted.

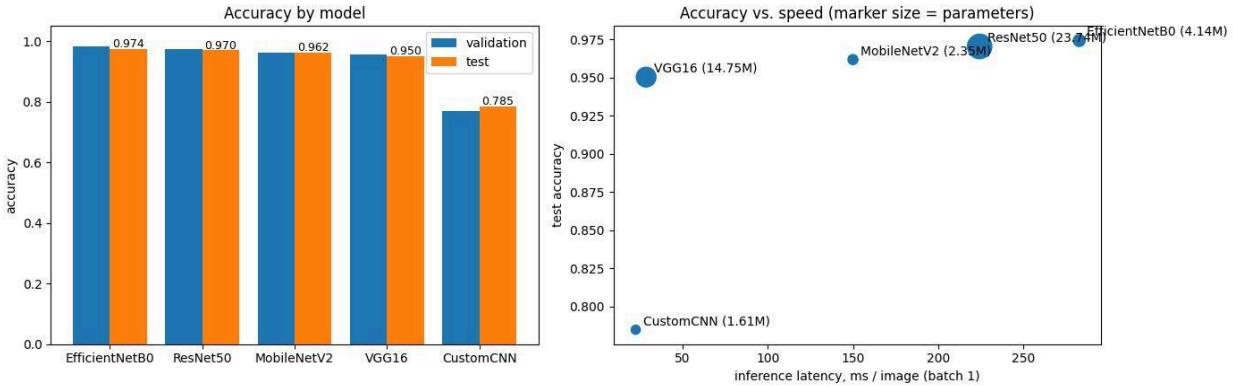


Figure 1. Validation and test accuracy by model (left), and test accuracy against inference latency (right).

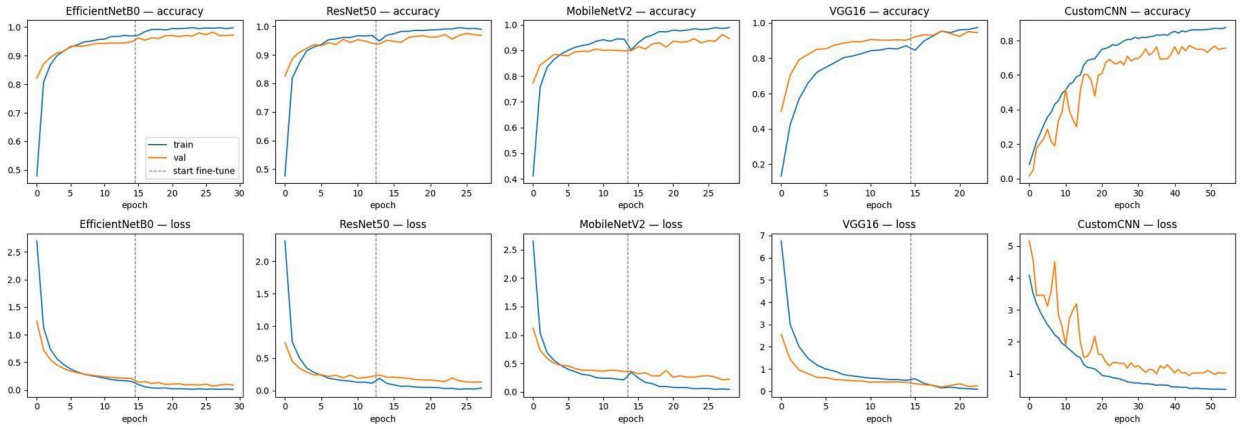


Figure 2. Training and validation curves. The dashed line marks the start of fine tuning (Stage 2).

Fine tuning improved every pretrained model. From the end of Stage 1 to the end of Stage 2, validation accuracy rose from 90.5% to 96.2% for MobileNetV2 (up 5.6 points), from 90.6% to 95.5% for VGG16 (up 4.9), from 94.8% to 98.2% for EfficientNetB0 (up 3.4) and from 95.4% to 97.5% for ResNet50 (up 2.1).

3.4 Analysis

Transfer learning versus training from scratch. All four pretrained models reach 95.0% to 97.4% test accuracy, while the custom CNN reaches 78.5%. The reason is the amount of data. With about 67 training photographs per object, a network trained from scratch must learn low level features such as edges and textures as well as the differences between objects. The pretrained backbones already learned general visual features from 1.2 million ImageNet images. The custom CNN was also unstable in the first 20 epochs, overfitted more (training accuracy near 87.5% against validation accuracy near 75%) and needed the longest training time (23.2 minutes).

Pretrained models. The four pretrained models are within 2.4 points of each other, so efficiency matters. EfficientNetB0 is the most accurate and has only 4.14 million parameters. ResNet50 is almost as accurate but about 5.7 times larger and took 1.7 times longer to train. VGG16 is the least accurate pretrained model despite 14.75 million parameters. MobileNetV2 is the smallest and fastest to train while still reaching 96.2%. All pretrained models overfit only mildly.

3.5 Error analysis of the best model

EfficientNetB0 misclassified 27 of the 1,045 test photographs (2.6% error). The confusion matrix (Figure 3) has a strong diagonal, and no pair of objects is confused more than twice. The weakest class is OBJ061 (recall 0.80). OBJ013 has the lowest precision (0.72). Several errors involve OBJ013, OBJ059 and OBJ061 being mistaken for one another, which suggests these objects share visual features. Figure 4 shows that in most errors the object is very small in the frame, partly hidden, in a dark scene or surrounded by clutter. Most wrong predictions also have low confidence (many between 0.33 and 0.63).

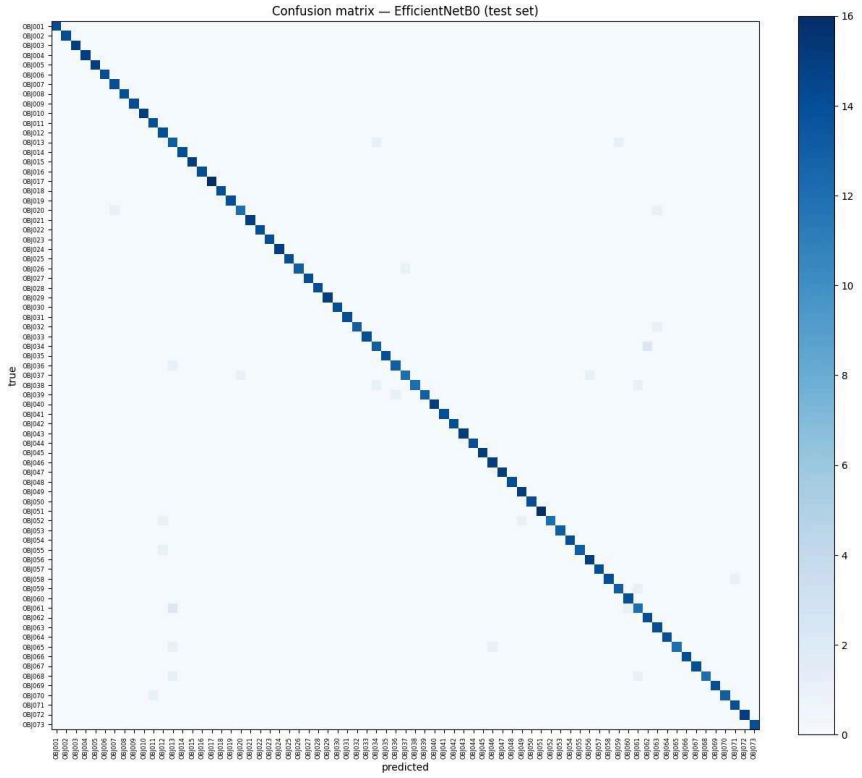


Figure 3. Confusion matrix of EfficientNetB0 on the Milestone 1 test set.

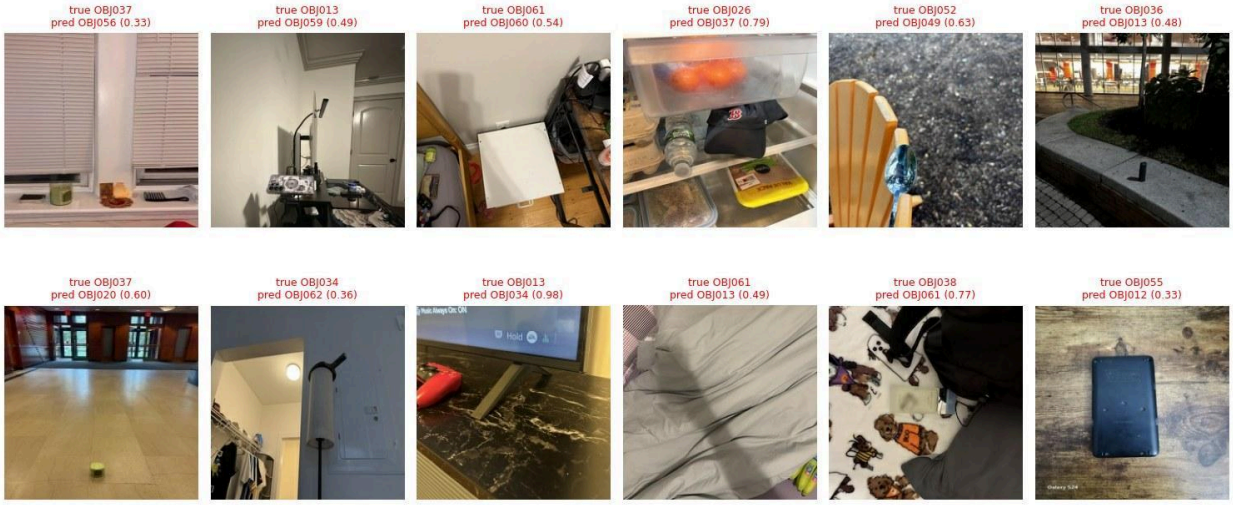


Figure 4. Examples of misclassified test photographs (true label, predicted label and confidence).

3.6 Milestone 1 model selection

We selected EfficientNetB0 with two stage fine tuning. It has the highest validation accuracy (98.2%), and also the highest test accuracy (97.4%), macro F1 (0.974) and top 5 accuracy (99.7%), with a small model (16.6 MB) and under 10 minutes of training. However, a classifier returns only one label per image and cannot say where the object is. The next task needs every object in an image and its location, so

Milestone 2 moves to object detection. The main lesson, that transfer learning clearly outperforms training from scratch on a dataset of this size, carries over.

4. Milestone 2: Building the Multi Object Detection Dataset

Following the instructors' clarifications, a multi object image is a grid of 224 × 224 photographs placed side by side, from 2 × 2 up to 5 × 5. Background only photographs are not used because grid images have no background. The detector must return the Object ID and bounding box of every object in the grid.

4.1 Bounding boxes for the single photographs

The shared dataset provides class labels but no bounding boxes. Each photograph shows one object, so it needs one box. We created the boxes automatically with two different open vocabulary detectors, YOLO World v2 (x) and Grounding DINO (tiny). Both were prompted with the object name from the class sheet (for example "Pencil Case" for OBJ036) and returned up to three candidate boxes per photograph. Comparing the two models catches many mistakes that either model makes alone.

A box is accepted when the two models agree, meaning that their boxes overlap with IoU ≥ 0.5; the final box is the average of the two. When the models do not agree, a single model's box is accepted only if that model is confident (Grounding DINO score ≥ 0.35 or YOLO World score ≥ 0.20). The remaining photographs are left out of the grid images, but never more than 20% of any Object ID: when more photographs were unreliable, the highest scoring boxes were kept ("rescued"). Table 5 shows the outcome. Labelling took 15.2 minutes with YOLO World and 26.7 minutes with Grounding DINO on a T4 GPU.

Label status	Rule	Photographs
Both models agree	Top boxes overlap with IoU ≥ 0.5	4,712 (67.8%)
Agree on 2nd or 3rd candidate	A lower ranked candidate matches (IoU ≥ 0.5)	568 (8.2%)
Grounding DINO only	No agreement, score ≥ 0.35	1,180 (17.0%)
YOLO World only	No agreement, score ≥ 0.20	8 (0.1%)
Rescued	Kept to respect the 20% limit	130 (1.9%)
Left out	No reliable box from either model	349 (5.0%)
Total		6,947 (100%)

Table 5. Outcome of the automatic box labelling.

The two models agreed on 5,280 photographs (76.0%), and where they agreed their boxes overlapped with a mean IoU of 0.944. We checked the labels visually with figures generated by the notebook: a random sample of 48 photographs (Figure 5), 48 of the 1,667 photographs where the models did not agree (Figure 6), and all 95 photographs of OBJ036 (Figure 7). In the random sample, 38 of the 43 kept boxes are correct; the other 5 are on another item or much too large (for example OBJ042_090, where both models boxed a game controller instead of the mouth guard case). For OBJ036, 69 of the 76 kept boxes are correct; the other 7 cover the rug, the chair or another item, mostly from the "agree on 2nd or

3rd candidate" rule. Based on these checks, about one kept box in ten is wrong. These boxes were not corrected by hand before training.

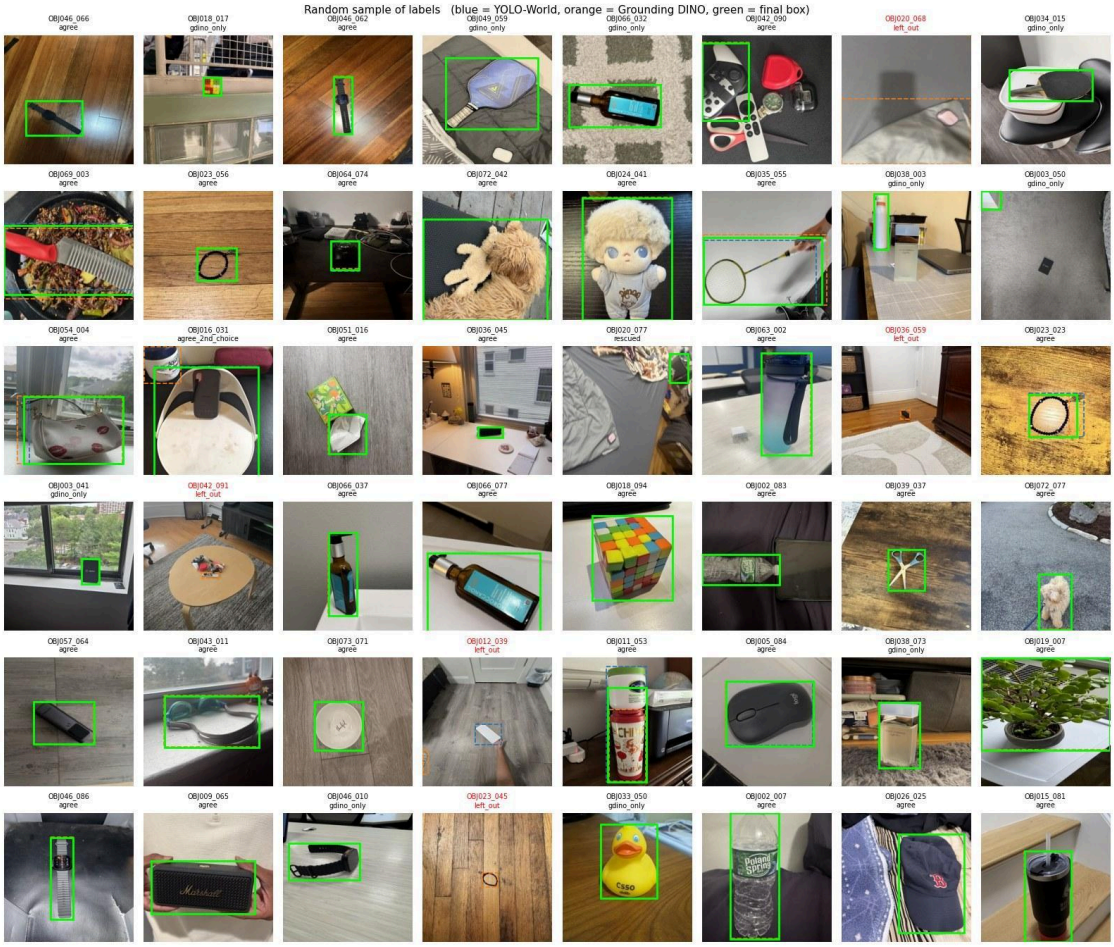


Figure 5. Random sample of 48 automatic labels (blue dashed = YOLO World, orange dashed = Grounding DINO, green = final box; red title = left out).

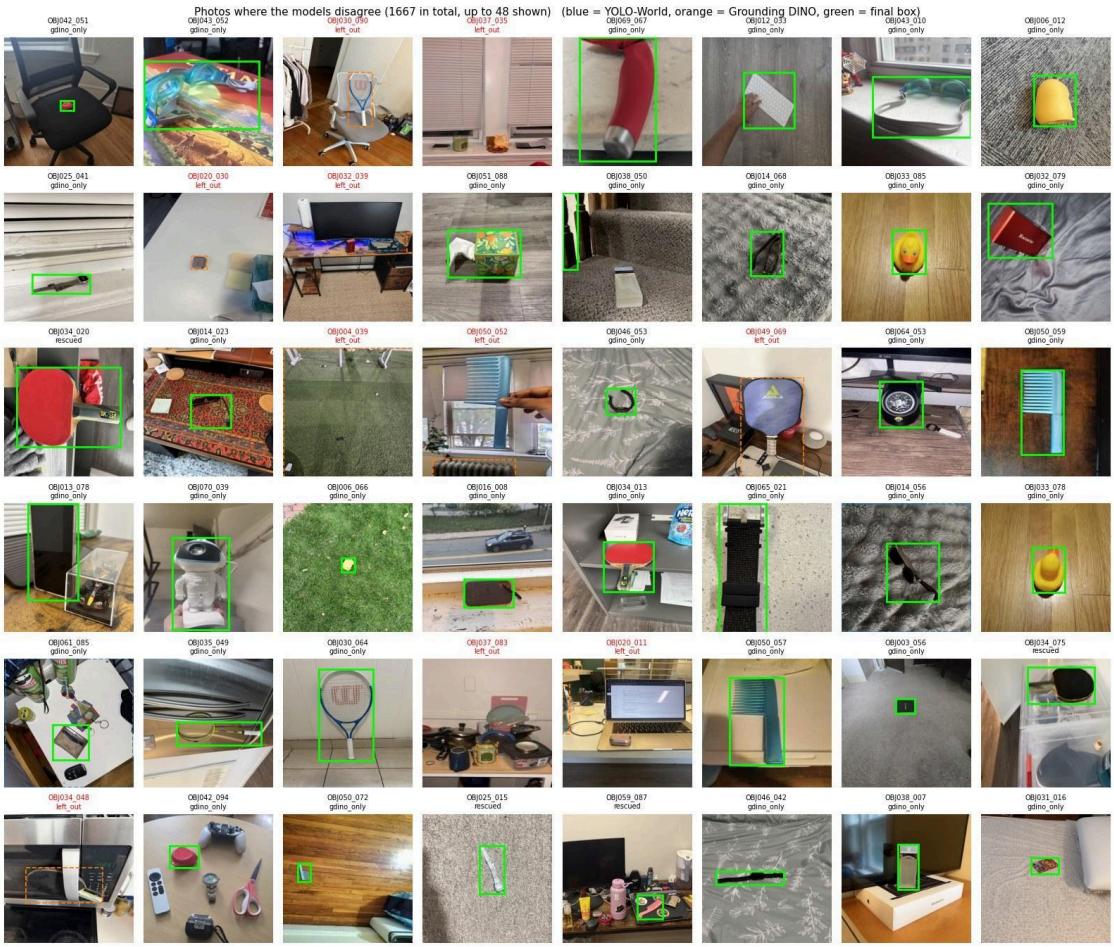


Figure 6. Sample of 48 photographs where the two labelling models did not agree.



Figure 7. All 95 OBJ036 photographs with their automatic labels, checked by hand.

In total, 349 photographs (5.0%) from 42 Object IDs were left out, in every case because neither model found the object with enough confidence. These are mostly photographs in which the object is very small or far away, dark on a dark background, or hidden among other items. Table 6 lists the number left out per Object ID. No Object ID exceeded the 20% limit, and the other 31 Object IDs had no photograph left out.

Object ID	Left out	Object ID	Left out	Object ID	Left out	Object ID	Left out
OBJ003	8	OBJ026	1	OBJ041	12	OBJ059	19
OBJ004	6	OBJ029	1	OBJ042	18	OBJ060	11
OBJ007	1	OBJ030	6	OBJ044	4	OBJ061	12

Object ID	Left out	Object ID	Left out	Object ID	Left out	Object ID	Left out
OBJ009	1	OBJ031	2	OBJ046	2	OBJ062	2
OBJ012	8	OBJ032	11	OBJ049	6	OBJ064	1
OBJ013	9	OBJ033	8	OBJ050	19	OBJ065	1
OBJ014	11	OBJ034	19	OBJ051	11	OBJ068	19
OBJ017	1	OBJ036	19	OBJ052	1	OBJ069	1
OBJ020	19	OBJ037	19	OBJ054	14	OBJ070	1
OBJ023	7	OBJ038	2	OBJ057	2		
OBJ025	18	OBJ040	15	OBJ058	1		

Table 6. Photographs left out per Object ID (only the 42 Object IDs with at least one).

4.2 Grid images

The multi object images are built by placing the 224 × 224 photographs side by side in the five grid sizes from the instructors' table. Training grids use only training photographs, validation grids only validation photographs and test grids only test photographs, and a check in the notebook confirms this. Within a split, photographs are drawn from a shuffled deck, so that each one is used about equally often and never twice in the same grid. Photographs that were left out in Section 4.1 are not used.

In the first Milestone 2 run we generated 800 training grids (160 per grid size). For the final model we increased this to 1,200 training grids (240 per grid size) to give the detector more training data. The validation and test grids (150 each) were not changed, so the results of the two runs can be compared directly. Table 7 shows the final grid dataset.

Grid	Objects per image	Image size	Train	Validation	Test
2 × 2	4	448 × 448	240	30	30
3 × 3	9	672 × 672	240	30	30
4 × 4	16	896 × 896	240	30	30
5 × 4	20	1120 × 896	240	30	30
5 × 5	25	1120 × 1120	240	30	30
Total images			1,200	150	150
Total objects			17,760	2,220	2,220

Table 7. Final grid dataset (1,500 multi object images, 22,200 objects).

For data augmentation, each tile of a training grid is randomly mirrored (with its box mirrored too) and receives random brightness, contrast and colour changes of up to ±25%. Validation and test grids are not augmented. YOLOv8 adds its own augmentation during training (mosaic, scaling, translation, HSV changes and flips). Each box is moved into grid coordinates and stored in YOLO format (class_id, x_center, y_center, width, height, normalised by the grid size). Figure 8 shows one example per grid size.



Figure 8. Example grid images with their true boxes, one for each grid size.

4.3 Lessons from our first composite pipeline

Our very first version pasted 2 to 5 resized photographs onto a 640 × 640 background photograph and used each whole pasted photograph as the box. It reached mAP@0.50 0.962 and mAP@0.50:0.95 0.962, but the two metrics were identical because the boxes only had to match the edges of the pasted squares, not the objects. That version also did not use grids, did not remove duplicate photographs (13 groups of identical photographs fell into different splits) and did not exclude OBJ054_017. The grid pipeline described above fixes all of these points. Its scores are lower but more meaningful, because the model must locate each object inside its cell.

5. Milestone 3: Final Model, Results and Demo

5.1 Training the final YOLOv8s model

A YOLOv8s detector initialised with COCO pretrained weights (11.2 million parameters) was fine tuned for the 73 Object IDs. The pretrained backbone and neck are reused, the detection head is replaced for 73 classes, and the whole network is trained on the grid images. Following the instructors' tip, the input size was set to 1120 px so that objects in 5 × 5 grids are not shrunk; at the default 640 px each 224 px cell would be reduced to about 128 px. The best epoch was chosen on the validation grids, and the test grids were used only once for the final results. Table 8 lists the settings.

Training setting	Milestone 2 run	Final model (Milestone 3)
Model	YOLOv8s, COCO pretrained	YOLOv8s, COCO pretrained (11.2 M parameters)
Training grids	800	1,200
Epochs	50 (all completed)	60 (all completed)
Input resolution	1120 × 1120	1120 × 1120
Batch size	8	8
Optimizer	AdamW, learning rate 0.00013 (chosen automatically)	AdamW, learning rate 0.00013 (chosen automatically)
Early stopping patience	15	15
Augmentation	Tile flips and colour changes; YOLOv8 defaults (mosaic off for last 10 epochs)	Same
Random seed	42	42

Training setting	Milestone 2 run	Final model (Milestone 3)
Hardware	Tesla T4 GPU	Tesla T4 GPU
Training time	41 min	56.6 min
Model file size	22.7 MB	22.7 MB

Table 8. YOLOv8s training configuration for the Milestone 2 run and the final model.

Figure 9 shows the training losses and the validation metrics for every epoch. The classification loss falls from 4.05 in the first epoch to 0.23 in the last. Validation mAP@0.50 passes 0.90 at epoch 10 and then stays between 0.91 and 0.92, so longer training would bring little gain. The selected checkpoint reaches precision 0.921, recall 0.890, mAP@0.50 0.920 and mAP@0.50:0.95 0.851 on the validation grids.

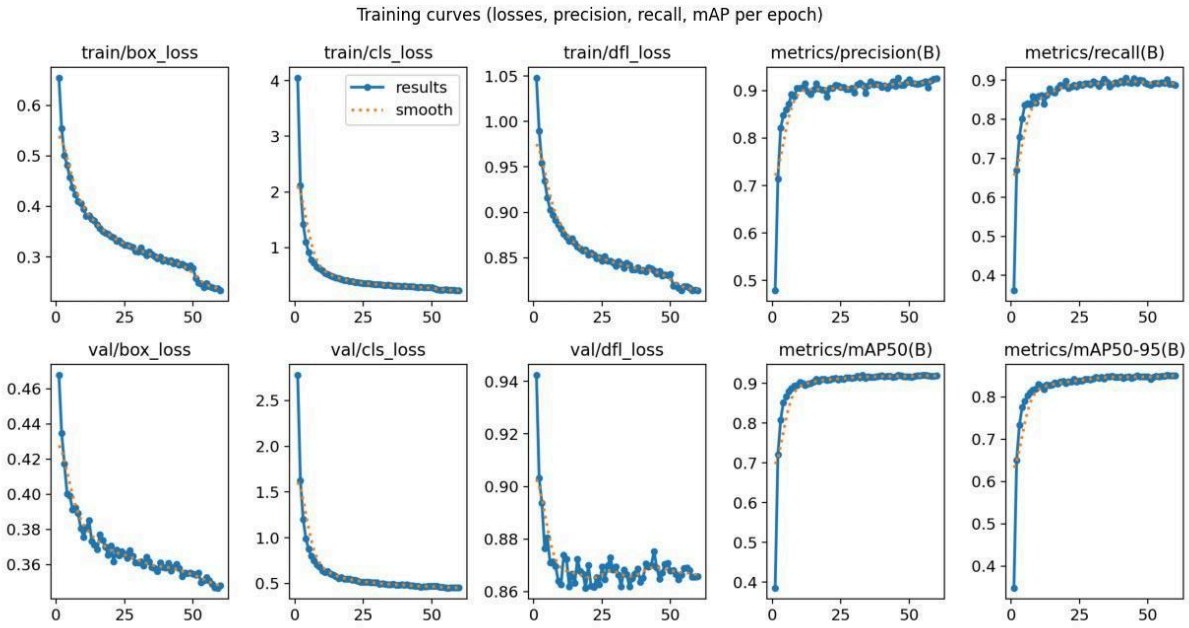


Figure 9. Training curves of the final model (losses, precision, recall and mAP per epoch).

5.2 Test results

The selected checkpoint was evaluated on the 150 held out test grids (2,220 objects), which are built only from test photographs. Precision and recall from Ultralytics are measured at the confidence threshold with the best F1 score. Table 9 compares validation and test.

Metric	Validation	Test
Precision	0.921	0.919
Recall	0.890	0.883
mAP@0.50	0.920	0.912
mAP@0.50:0.95	0.851	0.856

Table 9. Final YOLOv8s detection performance on the validation and test grids.

The test results match the validation results, so the model is not overfitted to the validation grids used to choose the checkpoint. Because each box surrounds the object inside its photograph rather than the whole photograph, mAP@0.50:0.95 (0.856) is lower than mAP@0.50 (0.912): the model has to place each box accurately, not only find the grid cell.

Grid	Test images	Objects	Precision	Recall	mAP@0.50	mAP@0.50:0.95
2 × 2	30	120	0.885	0.861	0.954	0.902
3 × 3	30	270	0.895	0.872	0.927	0.881
4 × 4	30	480	0.919	0.882	0.923	0.875
5 × 4	30	600	0.919	0.862	0.908	0.855
5 × 5	30	750	0.912	0.890	0.911	0.855
All	150	2,220	0.919	0.883	0.912	0.856

Table 10. Final model test results per grid size.

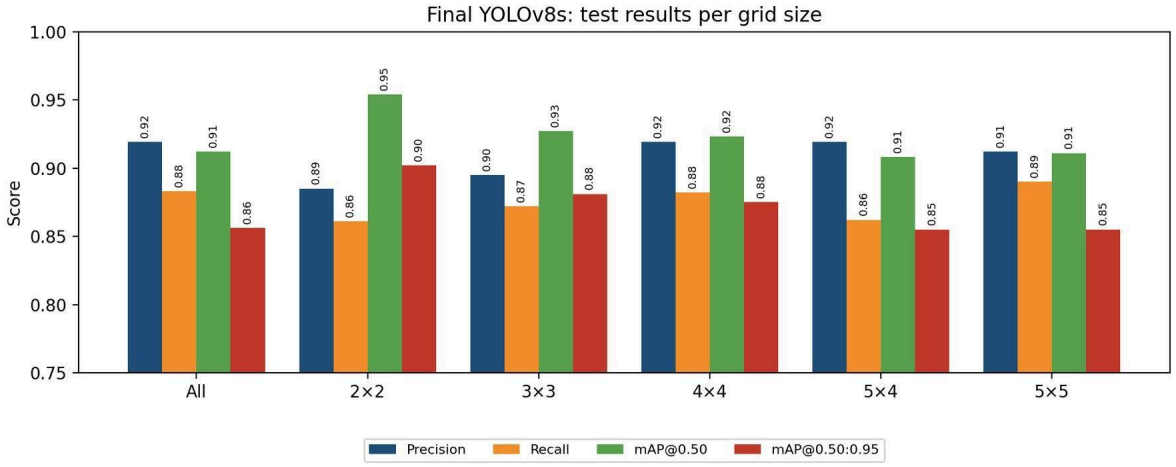


Figure 10. Test precision, recall, mAP@0.50 and mAP@0.50:0.95 for each grid size.

Performance is similar for all grid sizes. mAP@0.50 ranges from 0.908 (5 × 4) to 0.954 (2 × 2), so the model handles 25 objects per image almost as well as 4. The 2 × 2 grids contain only 120 test objects, so their precision and recall are the least certain.

5.3 Results at a fixed confidence threshold of 0.5

To measure the system as it would be used, we also ran the detector on every test grid with a fixed confidence threshold of 0.5. A detection counts as correct when the Object ID is right and the box overlaps the true box with IoU ≥ 0.5 (Table 11). At this threshold, 88.2% of the test objects are found with the correct ID and location, and 92.6% of the reported detections are correct. 25 of the 150 test grids (16.7%) are completely correct. Small grids are much easier to get fully correct (16 of 30 for 2 × 2), because a single miss among up to 25 objects is enough to fail a grid (0 of 30 for 5 × 5).

Grid	Objects	Correct	Wrong ID or extra	Missed	Grids fully correct	Precision	Recall
2 × 2	120	102	6	18	16 of 30	0.944	0.850
3 × 3	270	237	23	33	6 of 30	0.912	0.878
4 × 4	480	425	38	55	2 of 30	0.918	0.885
5 × 4	600	525	37	75	1 of 30	0.934	0.875
5 × 5	750	669	52	81	0 of 30	0.928	0.892
All	2,220	1,958	156	262	25 of 150	0.926	0.882

Table 11. Detection results on all test grids at a fixed confidence threshold of 0.5.

5.4 Results per Object ID

Performance varies by Object ID. Table 12 lists the ten weakest Object IDs on the test grids. Several of them are visually similar to another object (OBJ065 Analog Watch and OBJ025 Watch, OBJ014 Sunglasses and OBJ045 Wayfarer Sunglasses) or are often photographed small, dark or partly hidden. Many of them also had 11 to 19 photographs left out during labelling, so the remaining labels for these Object IDs are less reliable.

Object ID	Object	Precision	Recall	AP@0.50	AP@0.50:0.95
OBJ040	Clutcher	0.642	0.573	0.630	0.621
OBJ010	Nike Sneakers	0.606	0.600	0.656	0.526
OBJ065	Analog Watch	0.696	0.618	0.663	0.554
OBJ068	Neck Pillow	0.793	0.615	0.672	0.571
OBJ051	Tissue	0.691	0.639	0.704	0.543
OBJ014	Sunglasses	0.760	0.691	0.712	0.614
OBJ034	Table Tennis Racket	0.913	0.458	0.727	0.650
OBJ042	Mouth Guard Case	0.884	0.613	0.746	0.739
OBJ013	Mobile Phone	0.929	0.645	0.762	0.740
OBJ012	Computer Keyboard	0.773	0.704	0.773	0.725

Table 12. The ten weakest Object IDs on the test grids (sorted by AP@0.50).

Our group's objects perform well (Table 13). OBJ036 (Pencil Case) reaches precision 0.88, recall 0.90 and AP@0.50 0.887 on the test grids, and its AP@0.50:0.95 is also 0.887, which means its boxes are placed very accurately. OBJ038 (Perfume Bottle) reaches AP@0.50 0.882.

Object ID	Object	Test images	Objects	Precision	Recall	AP@0.50	AP@0.50:0.95
OBJ036	Pencil Case	19	20	0.880	0.900	0.887	0.887
OBJ038	Perfume Bottle	29	30	0.828	0.833	0.882	0.806

Table 13. Test results for our group's objects.

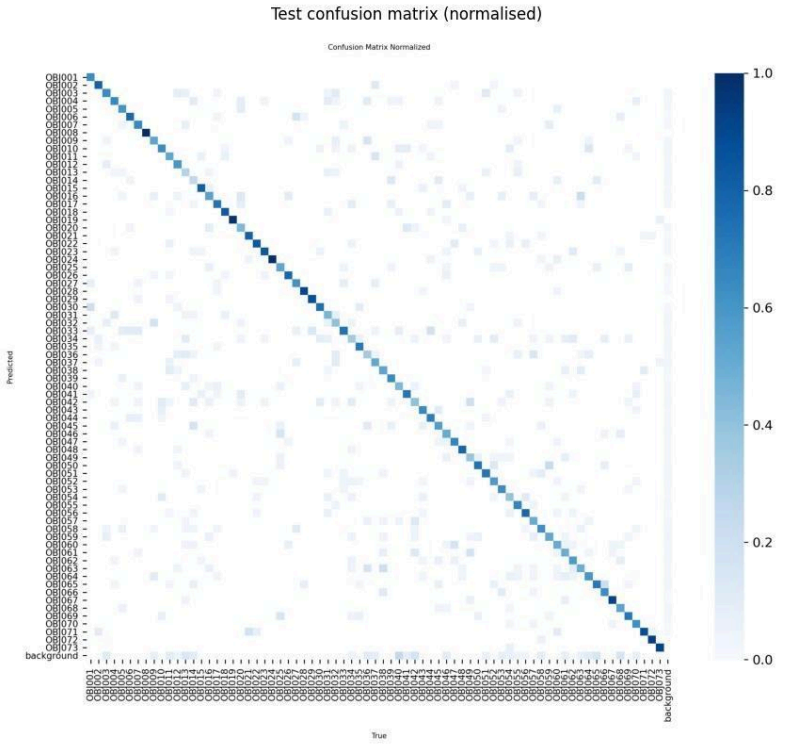


Figure 11. Normalised confusion matrix of the final model on the test grids.

The confusion matrix (Figure 11) has a strong diagonal. Almost all errors are missed objects (the "background" column) rather than wrong Object IDs: at confidence 0.5 the most frequent confusions occur only twice (OBJ003 predicted as OBJ012, and OBJ060 predicted as OBJ014).

5.5 Detection examples

Figure 12 shows four test grids that contain OBJ036 (3 × 3, 4 × 4, 5 × 4 and 5 × 5). OBJ036 is detected in all four with high confidence. Over the four grids, 64 of the 70 objects are detected with the correct ID and location, and the model makes 4 wrong detections.

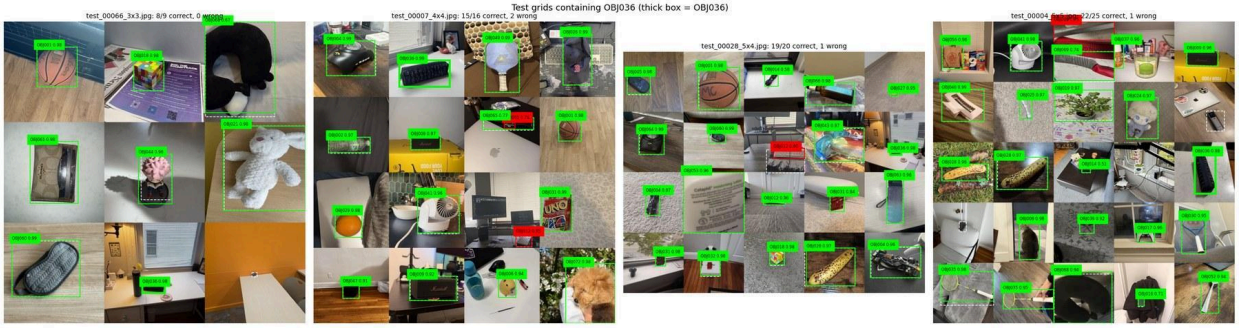


Figure 12. Example detections on test grids containing OBJ036 (white dashed = true box, green = correct, red = wrong; thick box = OBJ036).

Table 14 lists selected results for the 5 × 5 grid test_00004_5x5.jpg. The model finds 22 of the 25 objects with the correct ID, and its boxes match the true boxes within a few pixels, including OBJ036 in row 3, column 5 (confidence 0.98). It misses three objects (OBJ057, OBJ011 and OBJ060), which are among the smallest in the image because they were photographed from far away. The one wrong detection is a second OBJ069 box in the same cell. Compared with the Milestone 2 run, the final model now finds OBJ014 (confidence 0.51), which was missed before, and finds OBJ016 with higher confidence (0.71 instead of 0.53).

Object ID	Row, column	Confidence	Predicted (x1, y1, x2, y2)	True (x1, y1, x2, y2)
OBJ036	3, 5	0.98	(972, 488, 1048, 639)	(972, 488, 1048, 639)
OBJ019	2, 3	0.97	(451, 254, 668, 369)	(450, 254, 663, 367)
OBJ014	3, 3	0.51	(551, 536, 584, 560)	(551, 536, 584, 560)
OBJ016	5, 4	0.71	(761, 1018, 783, 1056)	(761, 1017, 783, 1057)
OBJ069 (extra)	1, 3	0.75	(448, 0, 672, 69)	no true object here
OBJ057	2, 5	missed	not detected	(1014, 330, 1081, 406)
OBJ011	3, 4	missed	not detected	(748, 528, 782, 565)
OBJ060	4, 1	missed	not detected	(95, 743, 123, 779)

Table 14. Selected detections in test_00004_5x5.jpg (22 of 25 objects correct, 1 wrong detection).

5.6 Error analysis

Figure 13 shows examples of test objects that were missed or wrongly identified at confidence 0.5. The same causes as in Milestone 1 appear: the object is very small in its cell, dark on a dark background, partly hidden, or seen from an unusual angle. Small objects are the main failure: objects photographed from far away occupy only about 20 to 35 px of a 224 px cell. The Object IDs with the most errors on the test grids are OBJ051 (15), OBJ065 (13), OBJ013 (12), OBJ034 (12), OBJ061 (11) and OBJ068 (11).

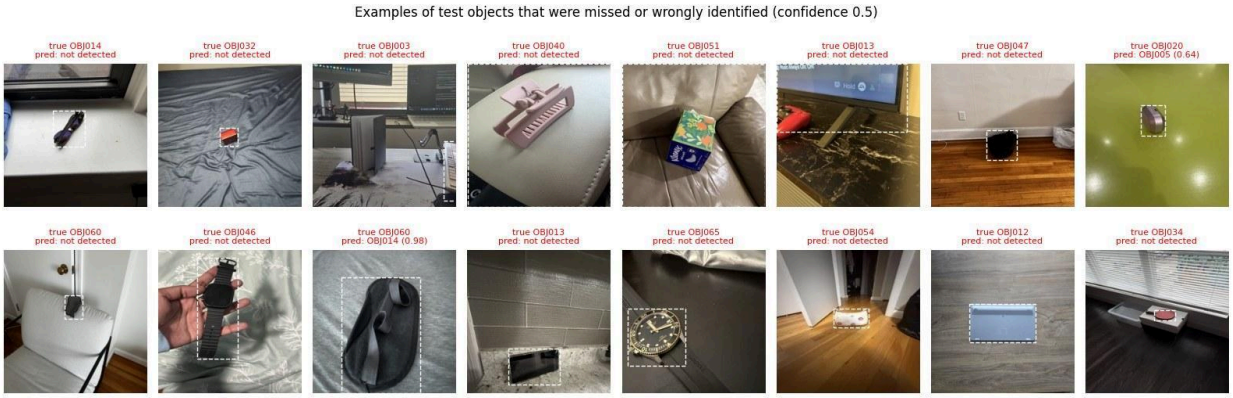


Figure 13. Examples of test objects that were missed or wrongly identified (confidence 0.5).

5.7 Live demo

Section 3.4 of the notebook is a standalone cell for the live demo. It loads the saved final weights and asks the user to upload images. A photograph of 336 px or smaller is treated as one 224 px photograph

and run at 448 px (single object task). Anything larger is treated as a multi object grid and run at 1120 px. For each image the cell draws the detections and prints a table with every object's ID, confidence, box (x1, y1, x2, y2) and grid row and column at confidence 0.5, and saves the table as a CSV file. Table 15 and Figure 14 show a test run of the demo cell.

Image	Grid	Objects in grid	Objects found
test_00002_4x4.jpg	4 × 4	16	15
test_00010_2x2.jpg	2 × 2	4	4
test_00015_2x2.jpg	2 × 2	4	4
test_00020_2x2.jpg	2 × 2	4	2
test_00051_3x3.jpg	3 × 3	9	9

Table 15. Test run of the live demo cell (confidence 0.5).

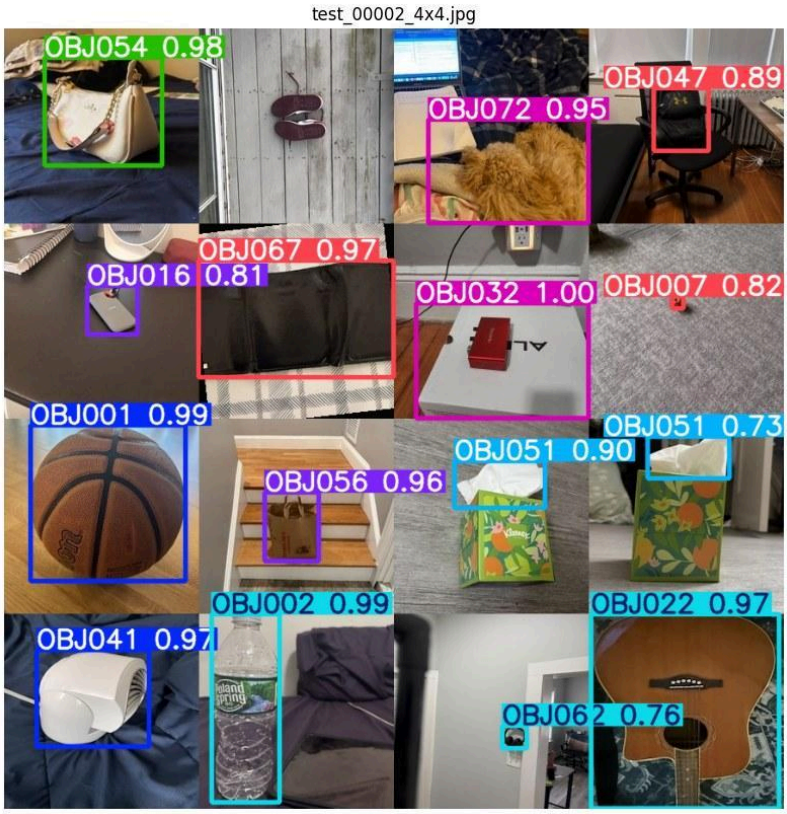


Figure 14. Live demo output for test_00002_4x4.jpg (15 objects found).

6. Comparison Across Milestones

6.1 Milestone 1 classifier versus final detector

Milestone 1 selected EfficientNetB0 for single object classification (97.4% test accuracy). The final model uses YOLOv8s to both identify and locate many objects in one image. Classification accuracy and detection mAP are different measures and should not be compared directly. Both use pretrained

networks (ImageNet for the classifier, COCO for the detector), the same 73 Object IDs and a stratified 70 / 15 / 15 split with seed 42. In both, the errors come mostly from objects that are small, dark, partly hidden or similar to another object.

6.2 Milestone 2 run versus final model

The Milestone 2 run and the final model were evaluated on the same 150 test grids, so their results can be compared directly (Table 16). With 50% more training grids and 10 more epochs, the final model improves precision, both mAP values and every measure at confidence 0.5. It finds 38 more objects correctly, misses 38 fewer and makes 6 fewer wrong detections. Recall at the best F1 threshold is slightly lower (0.883 against 0.896), but recall at the fixed threshold of 0.5 is higher (0.882 against 0.865).

Measure (test grids)	Milestone 2 run (800 grids)	Final model (1,200 grids)	Change
Precision	0.902	0.919	up 0.017
Recall	0.896	0.883	down 0.013
mAP@0.50	0.909	0.912	up 0.003
mAP@0.50:0.95	0.850	0.856	up 0.006
Correct detections at 0.5	1,920	1,958	38 more
Wrong ID or extra at 0.5	162	156	6 fewer
Missed objects at 0.5	300	262	38 fewer
Grids fully correct at 0.5	22 of 150	25 of 150	3 more
Training time	41 min	56.6 min	15.6 min longer

Table 16. Milestone 2 run compared with the final model on the same test grids.

The weakest Object IDs also improved. In the Milestone 2 run, OBJ050 (Hairbrush) was the weakest Object ID with a validation mAP@0.50 of 0.26; with the final model it reaches 0.826 on the test grids. The weakest test Object ID of the final model, OBJ040, reaches 0.630.

7. Final Model Selection and Justification

YOLOv8s fine tuned on 1,200 grid images is our final model. EfficientNetB0 is very accurate, but it gives only one Object ID per image and cannot say where an object is. YOLOv8s does both demo tasks with one model: it identifies a single object photograph, and it identifies and locates every object in a multi object image. It covers all 73 Object IDs, reaches 0.912 mAP@0.50 and 0.856 mAP@0.50:0.95 on 150 test grids, and keeps a similar level for every grid size up to 5 × 5. It is small (22.7 MB) and fast, with about 24 ms of inference per 1120 px image on a T4 GPU. We did not train other YOLO sizes or architectures on the same data, so we do not claim that YOLOv8s is the best possible choice; a larger model such as YOLOv8m is the natural next comparison.

8. Limitations

- The dataset is small (about 95 photographs per object). In Milestone 1 the test set has only 14 to 15 photographs per class, and in the final test grids each Object ID has only about 30 objects, so per class results are uncertain.
- The boxes were created automatically. Our checks suggest that about one kept box in ten is on another item or much too large; the same errors exist in the test labels, so the reported metrics include some label noise.
- 349 photographs (5.0%) were left out during labelling, mostly the hardest ones (very small, dark or cluttered), so the grids slightly under represent difficult photographs.
- Small objects are the main failure: objects photographed from far away occupy only about 20 to 35 px of a 224 px cell and are often missed.
- Grid cells do not overlap, so occlusion between objects is not tested, and all test images are synthetic grids built the same way as the training images.
- Each model was trained once with one seed and one split, and only one detector size (YOLOv8s) was evaluated.
- In Milestone 1, latency was measured with single image calls, which mostly reflects the number of layers rather than computation.

9. Next Steps

- Correct the wrong boxes found in the label checks (the notebook supports manual exclusions and manual boxes) and retrain.
- Compare YOLOv8s with a larger model such as YOLOv8m.
- Improve small object detection with scale augmentation or a higher input resolution.
- Test the model on real photographs that show several physical objects at once, with overlap between objects.
- Repeat training with different seeds to measure how stable the results are.

10. Conclusion

Across three milestones we built a system that recognises 73 objects photographed by the class. Milestone 1 showed that transfer learning clearly outperforms training from scratch on this small dataset, and EfficientNetB0 reached 97.4% test accuracy on single object photographs. Milestone 2 turned the task into multi object detection by creating bounding boxes with two open vocabulary detectors and building grid images from 2 × 2 to 5 × 5. Milestone 3 trained the final YOLOv8s detector on 1,200 grids. It reaches 0.912 mAP@0.50 and 0.856 mAP@0.50:0.95 on the held out test grids, finds 88.2% of the test objects with the correct ID and location at confidence 0.5, and works in the live demo for both single object and multi object images. The main remaining weakness is small, distant objects.